Team Details

Problem Statement: Unexpected machine failures cause downtime, repair costs, and production delays.

Theme: AI / Automation + Mechanical Engineering

Team Name: CODECRAFTERS

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ABOUT THE PROJECT

LifeLens AI — Detect Problems Before They Become Crises

Tagline: See the Signals. Prevent the Crisis.

Project Overview

LifeLens AI is an intelligent early-warning platform that analyzes multiple data sources to identify hidden patterns, detect unusual activities, and predict potential problems before they escalate into major crises.

Key Features

- **Multi-Source Analysis:** Combines maintenance reports, attendance trends, and infrastructure complaints.
- **Anomaly Detection:** Identifies unusual patterns and emerging issues.
- **Risk Prediction:** Estimates which problems require immediate attention.
- **Explainable Alerts:** Explains why an issue has been flagged.
- **Priority Dashboard:** Ranks problems by urgency and potential impact.

Primary Applications

- Campus safety and management
- Infrastructure maintenance
- Public service monitoring

Unique Value Proposition: Connects scattered information to uncover risks that conventional monitoring systems may overlook.

TECHNICAL APPROACH

1. Data Collection

- Maintenance requests and complaint records
- Attendance and operational trends
- Infrastructure inspection reports
- Historical incident records

2. Data Processing

- Clean and organize incoming data.
- Handle missing values and duplicate records.
- Anonymize personal information where appropriate.

3. AI Intelligence Engine

- Machine Learning for risk prediction
- Anomaly Detection for unusual patterns
- Natural Language Processing (NLP) to understand complaints
- Risk Scoring to prioritize emerging issues

4. Explainable Alert Generation

- Identify the factors contributing to each risk.
- Explain why an alert was generated.
- Recommend appropriate next steps.

5. Interactive Dashboard

- Real-time or periodically updated risk indicators
- High-, medium-, and low-priority alerts
- Historical trends and risk analysis
- Recommended actions and resolution tracking

Component	Technology
Frontend	React.js
Backend	Python + FastAPI
Machine Learning	Scikit-learn
NLP	Sentence Transformers
Database	PostgreSQL
Data Visualization	Recharts
AI Explainability	Feature importance and rule-based explanations

FEASIBILITY AND VIABILITY

Technical Feasibility
Uses open-source Python and AI/ML technologies.
Supports multiple data sources and anomaly detection.
Operational Feasibility
Centralizes monitoring and prioritizes critical issues.
Integrates with existing workflows.
Economic Viability
Reduces development costs using open-source tools.
Improves resource allocation and helps prevent costly failures.

Phase	Activities
Phase 1	Collect and prepare sample datasets
Phase 2	Develop anomaly detection and risk scoring
Phase 3	Build the dashboard and alert system
Phase 4	Test predictions using historical or simulated data
Phase 5	Evaluate performance and prepare for pilot deployment

IMPACT AND BENEFITS

Expected Impact

1. Early Problem Detection

Identifies warning signs before minor issues develop into major disruptions.

2. Faster Decision-Making

Helps administrators understand which issues require immediate attention.

3. Improved Resource Allocation

Prioritizes maintenance teams, budgets, and operational resources based on risk.

4. Reduced Operational Disruptions

Supports proactive maintenance and timely issue resolution.

5. Explainable AI

Provides understandable reasons behind risk scores, helping users make informed decisions.

6. Scalable Intelligence

Supports multiple institutions and infrastructure environments through a common platform.

RESEARCH AND REFERENCES

1. **Anomaly Detection:** Chandola et al. (2009) — *Anomaly Detection: A Survey.*
2. **Explainable AI:** Ribeiro et al. (2016) — *Why Should I Trust You?*
3. **Machine Learning:** Scikit-learn
4. **Natural Language Processing:** Sentence Transformers
5. **Web Development:** FastAPI and React
6. **Data Privacy:** NIST Privacy Framework